COURSE WORK · NATIONAL POLYTECHNIC UNIVERSITY OF ARMENIA

House price prediction with Ridge regression

A regularized linear model trained with gradient descent, built from scratch in NumPy on the Ames Housing dataset

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STAGE 1 · PROBLEM STATEMENT

Predict the sale price of a house from its features

Supervised regression: the target is SalePrice in US dollars, the inputs are measurable properties of the house. A model like this supports valuation, flags over- or under-priced listings and shows which factors drive the price.

1

Study linear regression, Ridge/Lasso regularization and gradient descent

2

Select and prepare a subset of the Ames Housing data

3

Analyze feature–price relations with Pearson correlation

4

Implement Ridge with batch gradient descent using only NumPy

5

Evaluate with MAE, RMSE, R^2 and actual-vs-predicted plots

6

Draw conclusions and describe the planned Lasso algorithm

STAGE 2 · THEORY

Why regularization: Ridge (L2) vs Lasso (L1)

Plain least squares overfits when features are many or correlated (multicollinearity). A penalty on the weights keeps them small and stable.

	Ridge (L2)	Lasso (L1)
Penalty	$\alpha\|\|w\|\|_2^2$, sum of squares	$\alpha\|\|w\|\|_1$, sum of absolute values
Effect on weights	Shrinks all smoothly	Shrinks, some become exactly 0
Feature selection	No	Yes (sparse solution)
Optimization	Gradient descent, closed form exists	Proximal / soft-threshold steps

$$J(w,b) = 1/2N \cdot \Sigma(\hat{y} - y)^2 + \alpha/2N \cdot ||w||^2 \quad \partial J/\partial w = X^T e/N + \alpha \cdot w/N \quad w \leftarrow w - \eta \cdot \partial J/\partial w$$

FORMULAS USED IN THE PROJECT

From preprocessing to evaluation

Standardization

$$z = (x - \mu) / \sigma$$

Mean 0 and standard deviation 1 for the 7 numeric features

Pearson correlation

$$r = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \sqrt{(\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2)}$$

Linear link between a feature and SalePrice

Ridge loss

$$J = 1/2N \cdot \Sigma(\hat{y}_i - y_i)^2 + \alpha/2N \cdot \Sigma w_j^2$$

Prediction $\hat{y} = Xw + b$; bias b is not penalized

Gradients and update

$$\partial J/\partial w = X^T e/N + \alpha w/N$$

$$\partial J/\partial b = \Sigma e/N$$

$$w \leftarrow w - \eta \cdot \partial J/\partial w$$

$e = \hat{y} - y$

Error metrics

$$MAE = 1/N \cdot \Sigma|y_i - \hat{y}_i|$$

$$RMSE = \sqrt{(1/N \cdot \Sigma(y_i - \hat{y}_i)^2)}$$

Both in dollars; RMSE punishes big errors more

R^2 score

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2$$

Share of price variance explained by the model

STAGE 3 · DATASET

Ames Housing: 2,930 sales, 8 selected features

Loaded from Kaggle with kagglehub and pandas. The full table has 82 columns; a compact, interpretable subset was chosen by hand.

Feature	Type	Meaning
Overall Qual	Ordinal 1–10	Materials and finish quality
Gr Liv Area	Numeric	Above-ground living area, sq ft
Year Built	Numeric	Original construction year
Garage Cars	Numeric 0–5	Garage capacity in cars
Total Bsmt SF	Numeric	Basement area, sq ft
Bedroom AbvGr	Numeric 0–8	Bedrooms above ground
Lot Area	Numeric	Lot size, sq ft
Neighborhood	Categorical (28)	Location within Ames
SalePrice	Target	Sale price in USD

STAGE 4 · DATA PREPARATION

Four steps from raw table to model input

1

Missing values

Garage Cars and Total Bsmt SF had one gap each. The 2 rows (0.07%) were dropped with dropna(): 2,930 → 2,928 rows.

2

Train/test split

train_test_split, 80/20, random_state = 42: 2,342 training and 586 test houses, reproducible.

3

Scaling

$z = (x - \mu) / \sigma$ on the 7 numeric features, written by hand in pandas. SalePrice stays in dollars.

4

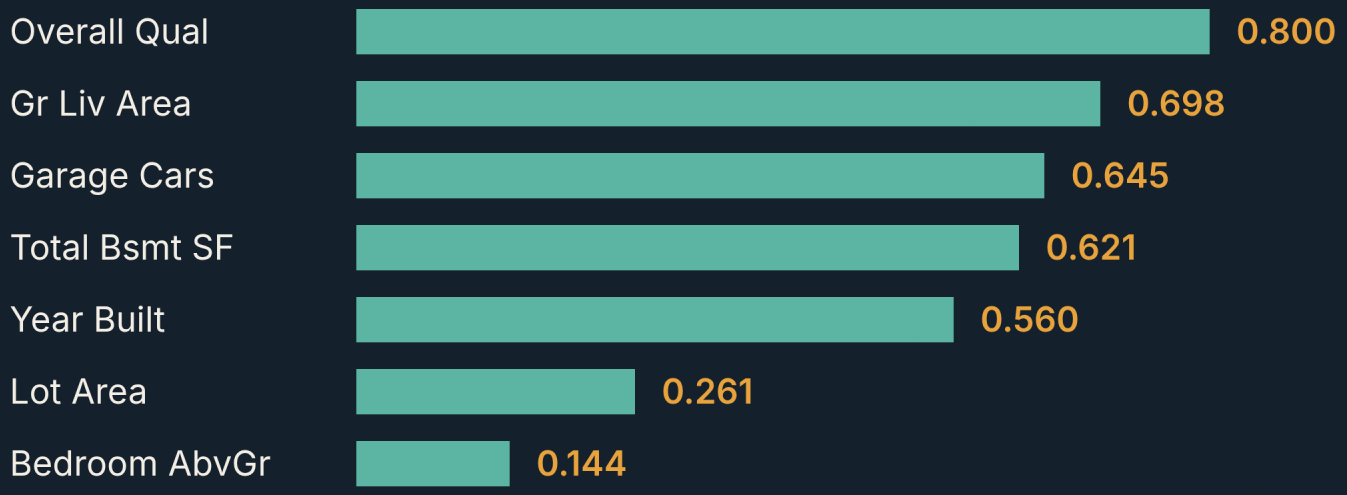
Encoding

OneHotEncoder(drop='first') fitted on training data only: 27 dummy columns, 34 features in total.

Checks before modelling: Overall Qual 1–10, Garage Cars 0–5, Bedroom AbvGr 0–8, 28 neighborhoods, no impossible values. Scaling puts all weights on one scale, so a single learning rate works and the penalty treats features fairly.

STAGE 5 · EXPLORATORY ANALYSIS

Quality and living area drive the price



Pearson r with SalePrice, training set, computed with an explicit loop

What it showed

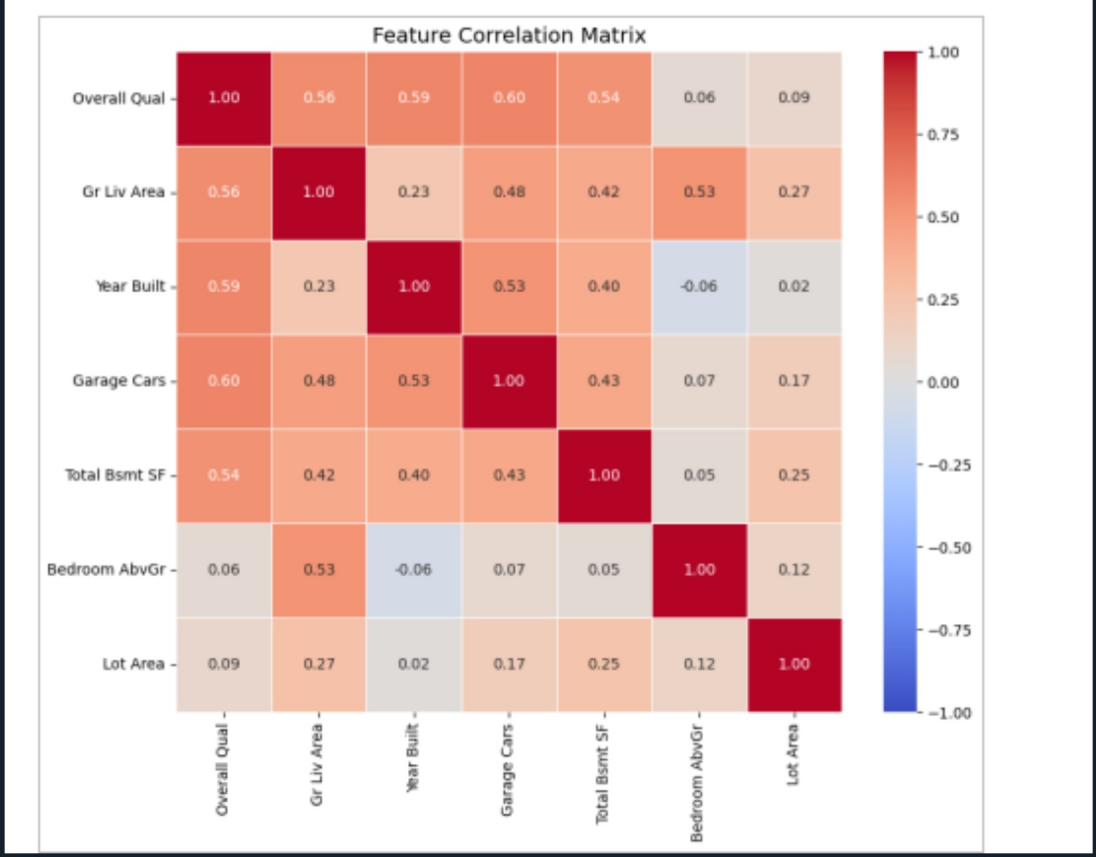
Bedrooms add little once living area is known (r between them 0.53).

The heatmap shows moderate multicollinearity: Overall Qual correlates 0.54–0.60 with four other features, and no pair exceeds 0.60.

That supports choosing a regularized model.

STAGE 5 · CORRELATION MATRIX

Feature correlation heatmap



How it was built

Pearson matrix of the 7 numeric training features, drawn with seaborn.

What it shows

Overall Qual is linked to Garage Cars (0.60), Year Built (0.59), Gr Liv Area (0.56) and Total Bsmt SF (0.54).

Bedrooms and living area: 0.53. No pair is above 0.60, so collinearity is moderate and Ridge fits well.

STAGE 6 · IMPLEMENTATION

Ridge regression with batch gradient descent

```
w = np.zeros(34); b = 0.0
for i in range(8000):
  e = X_train @ w + b - y_train
  db = e.sum() / N
  dw = X_train.T @ e / N + alpha / N * w
  b = b - lr * db
  w = w - lr * dw
```

Written directly from the Ridge equations, no ready-made regression classes. The L2 term is added to the weight gradient only.

Hyperparameter	Value
Learning rate η	0.001
Regularization α	10
Iterations	8,000
Start	w = 0, b = 0
Samples N	2,342

Metrics (MAE, RMSE, R^2) were also implemented by hand and printed every 1,000 iterations.

STAGE 7 · RESULTS

Training converges to $R^2 \approx 0.81$

Iteration	MAE ($)	RMSE ($)	R^2
0	180,142	195,878	−5.48
1,000	63,394	71,960	0.12
2,000	28,935	41,368	0.71
3,000	23,402	35,605	0.79
5,000	22,463	34,207	0.80
7,000	22,190	33,847	0.81

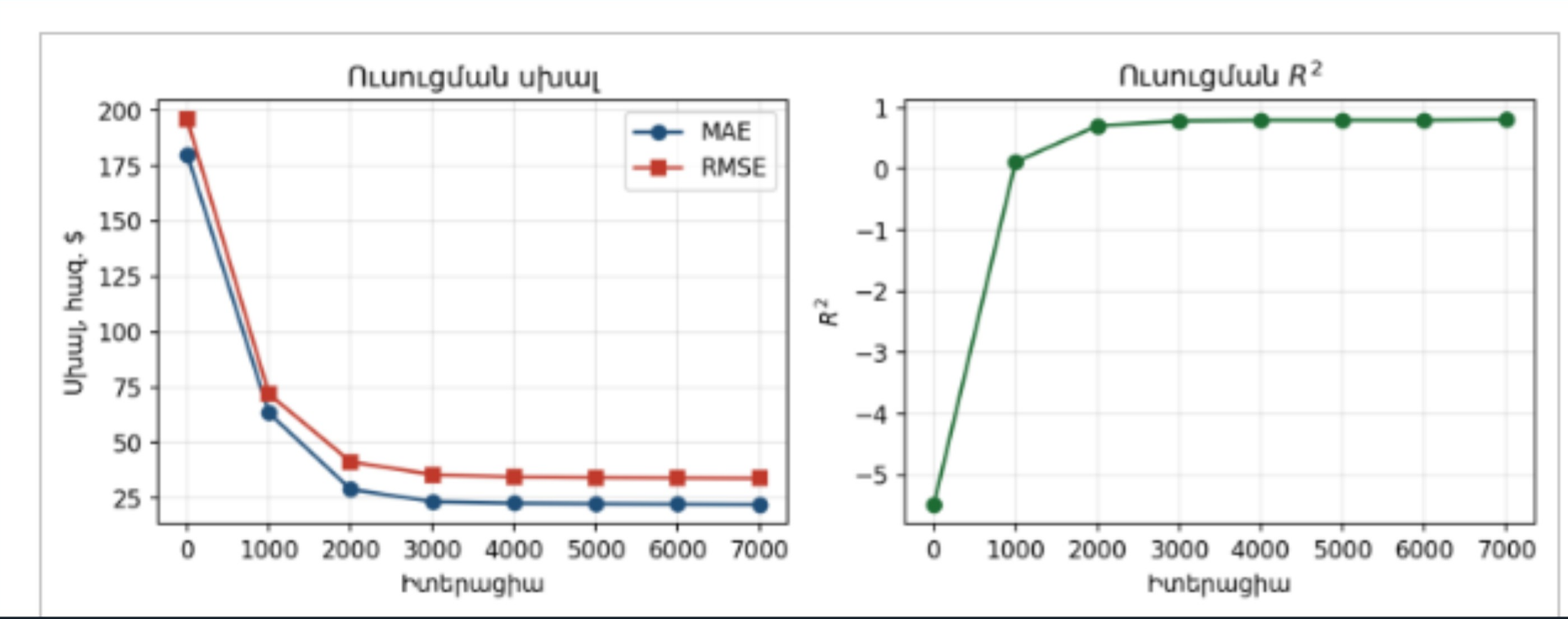
At start the model predicts 0 for every house, so MAE equals the mean price (about $180K).

Error falls fast in the first 3,000 steps, then improves slowly, the usual behavior near the minimum.

Final MAE is about 12% of the mean price. Test scatter looks like training scatter.

STAGE 7 · TRAINING CURVES

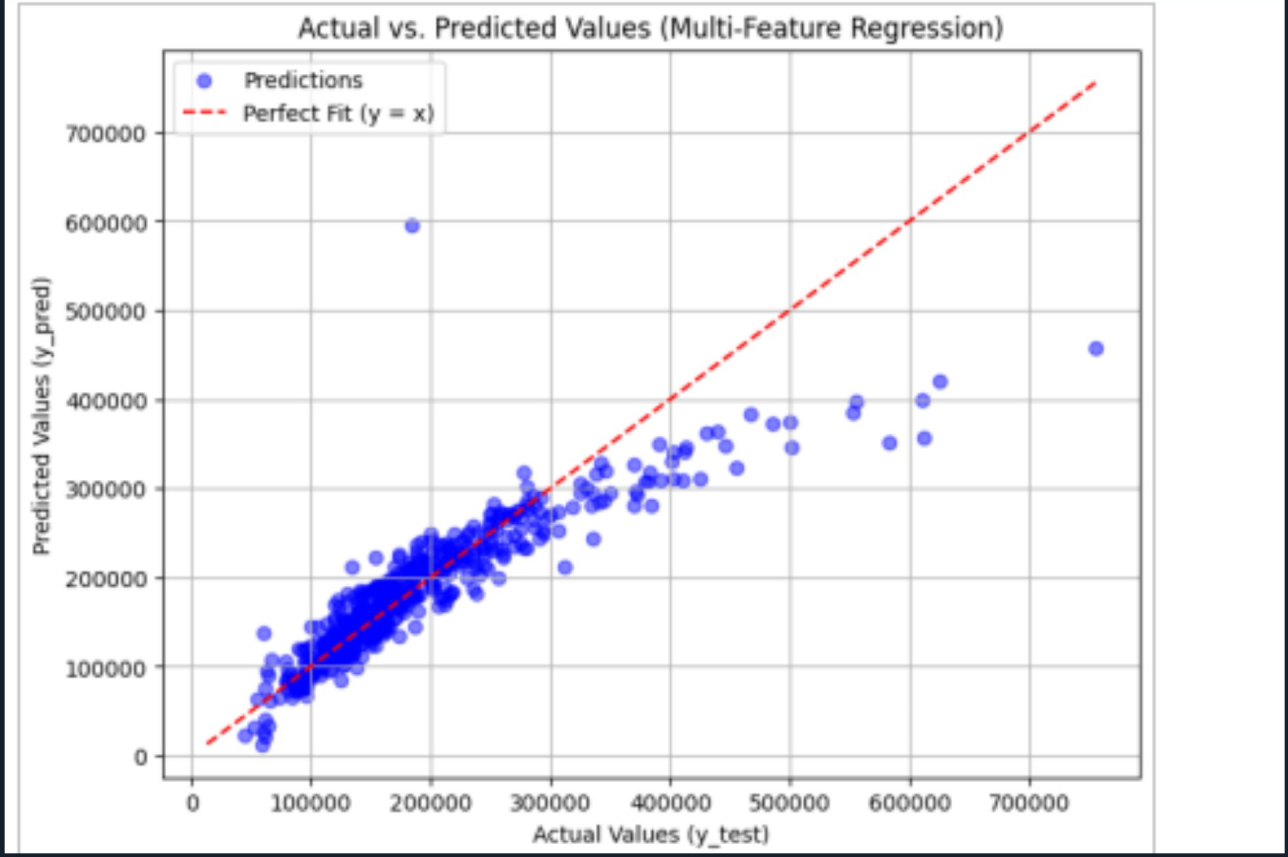
Gradient descent convergence



MAE, RMSE (left) and R^2 (right) on the training set, recorded every 1,000 iterations. Error drops sharply until about 3,000 steps, then flattens near $R^2 \approx 0.81$.

STAGE 7 · TEST-SET EVALUATION

Actual vs predicted prices



How to read it

Each dot is a test house; the red dashed line y = x is a perfect prediction.

What it shows

Most homes (about $50K–$350K) sit close to the line, and the spread resembles training, so no clear overfitting.

Homes above about $400K fall below the line (underestimated), and one house is predicted near $595K for an actual price near $185K.

STAGE 8 · ANALYSIS AND NEXT STEPS

Good on typical homes, with clear room to improve

What the plots showed

Predictions track the diagonal for homes of about $50K–$350K.

Homes above about $400K are underestimated.

A few large outliers inflate RMSE; some cheap homes get negative predictions.

Planned improvements

Report MAE, RMSE, R^2 on the test set.

Scale test data with training μ and σ.

Choose α by cross-validation; predict log(SalePrice).

Implement Lasso (soft-thresholding) and compare with Ridge.

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House price prediction with Ridge regression

Regularized linear model trained with gradient descent, written from scratch in NumPy

Data

Ames Housing: 2,928 homes, 8 features (7 numeric, 1 categorical), 34 columns after one-hot encoding

Preparation

80/20 split (2,342 train, 586 test), z-score scaling, Pearson correlation analysis

Key finding

Overall quality (r = 0.80) and living area (r = 0.70) are the strongest price predictors

Model

Batch gradient descent, learning rate 0.001, α = 10, about 8,000 iterations

0.81 — R^2, training set

$22.2K — MAE, about 12% of the mean price

$33.8K — RMSE, inflated by a few outliers

Test-set scatter matches training, so no overfitting. Limits: homes above $400K are underestimated; Lasso is described but not yet implemented.